

CSTE 5131

Machine Learning

Fateha Khanam Bappee, Ph.D.

Department of Computer Science and Telecommunications Engineering
Noakhali Science and Technology University
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**Courtesy: Professor Qigang Gao
Dalhousie University**



Classification: Basic Concepts

- Classification problem overview
- General issues of classification DM
- Mining classification model by decision tree induction
- Bayesian Classification
- Other classification methods
- Summary

Classification Problem Overview

E.g.1, Recap the classification problem: Risk Analysis Modeling

Target concept

NO.	RISK	CREDIT HISTORY	DEBT	COLLATERAL	INCOME
1.	high	bad	high	none	\$0 to \$15k
2.	high	unknown	high	none	\$15 to \$35k
3.	moderate	unknown	low	none	\$15 to \$35k
4.	high	unknown	low	none	\$0 to \$15k
5.	low	unknown	low	none	over \$35k
6.	low	unknown	low	adequate	over \$35k
7.	high	bad	low	none	\$0 to \$15k
8.	moderate	bad	low	adequate	over \$35k
9.	low	good	low	none	over \$35k
10.	low	good	high	adequate	over \$35k
11.	high	good	high	none	\$0 to \$15k
12.	moderate	good	high	none	\$15 to \$35k
13.	low	good	high	none	over \$35k
14.	high	bad	high	none	\$15 to \$35k

Data from credit history of loan applications

Classification Problem Overview

E.g.2, Learning a model for predicting
PlayTennis.

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rain	mild	high	weak	Yes
5	rain	cool	normal	weak	Yes
6	rain	cool	normal	strong	No
7	overcast	cool	normal	strong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rain	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	false	Yes
14	rain	mild	high	strong	No

Target concept

Classification Problem Overview

E.g.3, Learning a model to predict the relevance of search results on homedepot.com

- **Kaggle competition:** Predict the relevance of search results on homedepot.com
 - <https://www.kaggle.com/c/home-depot-product-search-relevance>
 - **Started:** Monday 18 January 2016
 - **Ends:** Monday 25 April 2016
- **Data fields** (<https://www.kaggle.com/c/home-depot-product-search-relevance/data>)
 - **id** - a unique Id field which represents a (search_term, product_uid) pair
 - **product_uid** - an id for the products
 - **product_title** - the product title
 - **product_description** - the text description of the product (may contain HTML content)
 - **search term** - the search query
 - **relevance** - the average of the relevance ratings for a given id (from 1 to 3)
 - **name** - an attribute name
 - **value** - the attribute's value

Target concept

Define Classification Data Mining

- General description/ definition:
 - Given a **training data set** $D = \{t_1, t_2, \dots, t_n\}$ which has n rows (records, tuples) and m columns (features, attributes); for a given **target concept** (i.e., a selected feature) with **classes** $C = \{c_1, \dots, c_p\}$, discover a **model** (knowledge/pattern/rules) for **predicting** a target value c_i of a new instance
- Classification DM: Find a mapping function $f: D \Rightarrow C$, i.e., the classification knowledge or model, where each t_i in D is assigned to a class c_j of C .

Example: Classification DM- find the model (coded in DT) about “Risk” (target concept)

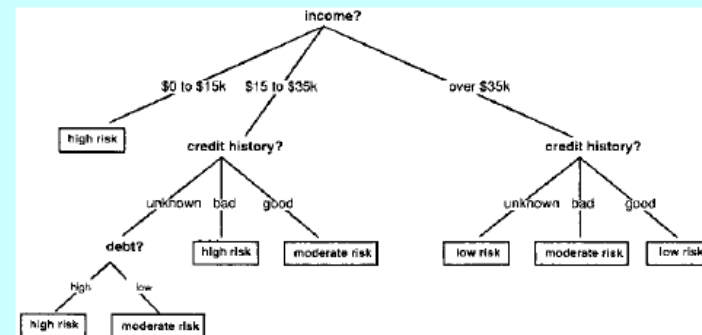
Training set: **D**

NO.	RISK	CREDIT HISTORY	DEBT	COLLATERAL	INCOME
1.	high	bad	high	none	\$0 to \$15k
2.	high	unknown	high	none	\$15 to \$35k
3.	moderate	unknown	low	none	\$15 to \$35k
4.	high	unknown	low	none	\$0 to \$15k
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Data from credit history of loan applications



Classification
Model: ***f***



A simplified decision tree for credit risk assessment.

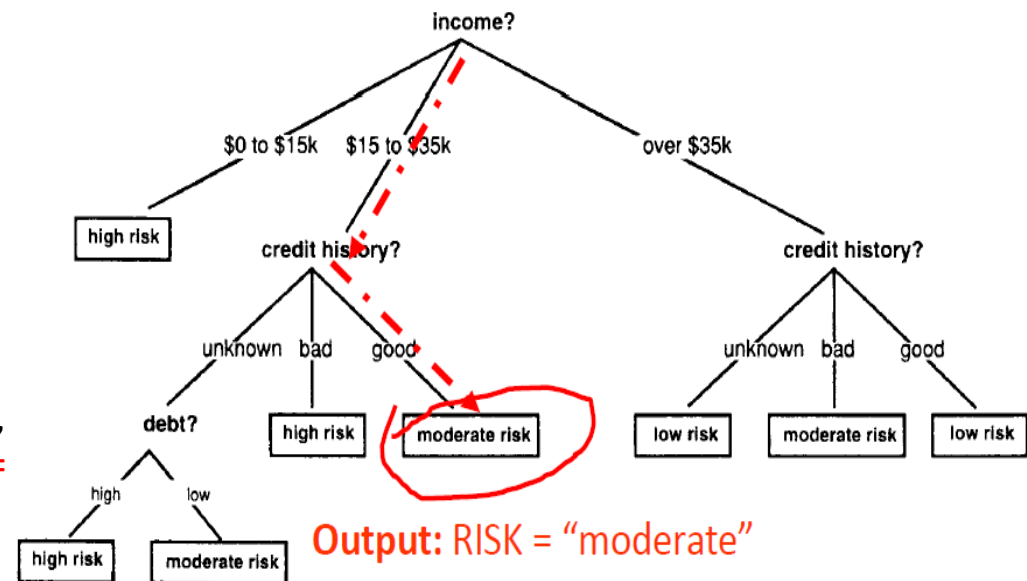
Apply f (coded as a classifier) for *Class Predication*:

- Predict the risk value for a new instance:

Input: <CREDIT_HIS=good, DEBT=low, COLLATERAL=unknown, INCOME=\$30k>

Output: (value of target attribute):
RISK = ?

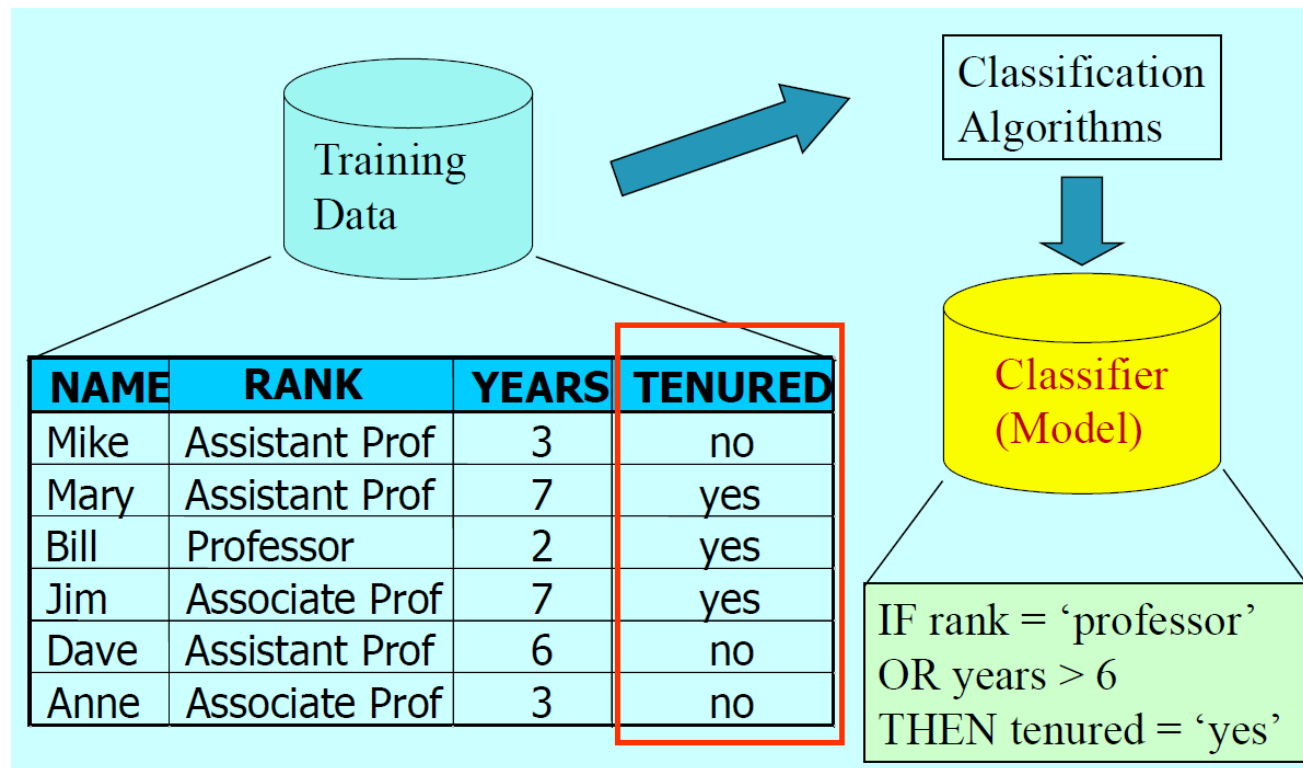
Decision rule: If INCOME=\$30k (i.e., 15-35k) and CREDIT_HIS=good, then Risk = moderate



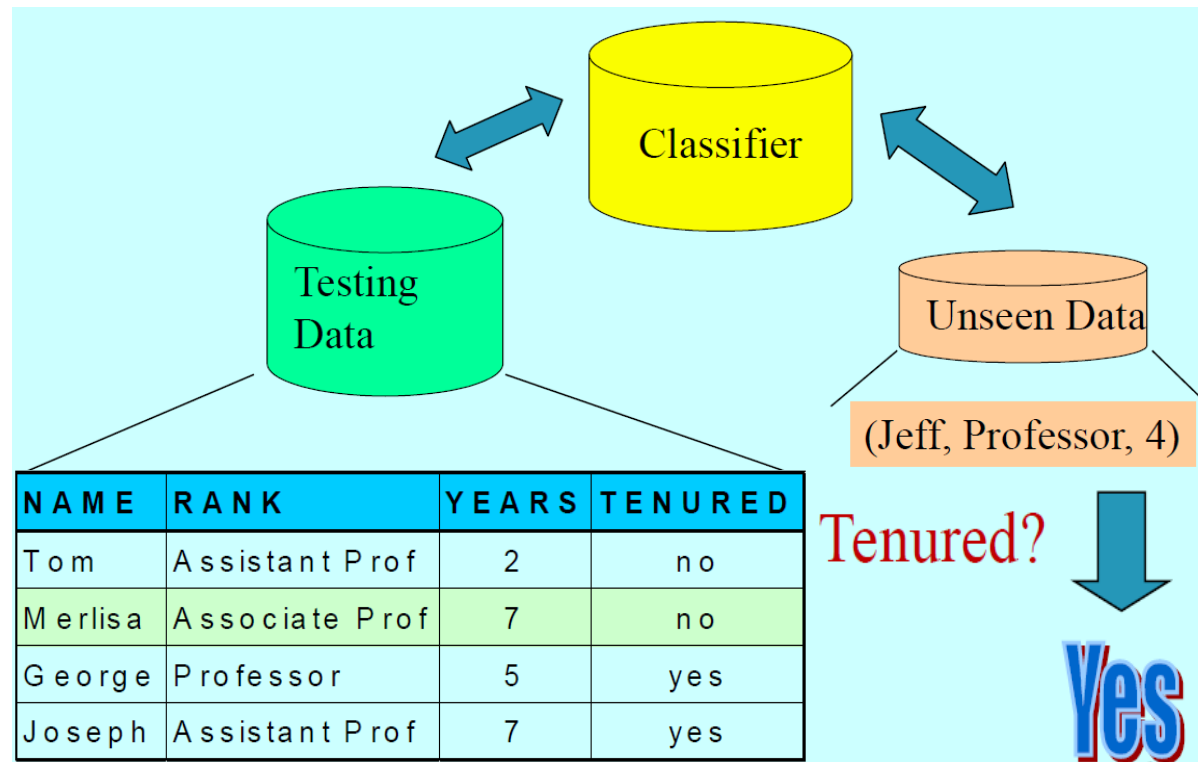
Output: RISK = "moderate"

A simplified decision tree for credit risk assessment.

Classification pattern discovery: model construction



Classification (prediction): class labeling





Classification application development: a three-phase process

- **Model construction:** find classification rules from the training data set

1. Training:

- Each tuple is assumed to belong to a predefined class, as determined by the class label attribute
- The set of tuples used for model construction is training set
- The model is represented as classification rules, decision trees, or distributed network weights, mathematical formula, etc.

- **Model usage:** for classifying unseen objects

2. Testing: Estimate accuracy of the model

- The known label of test sample is compared with the classified result from the model
- Accuracy rate is the percentage of test set samples that are correctly classified by the model
- Test set is independent of training set

3. Predicting: If the accuracy is acceptable, use the model to classify unseen data



Evaluating classification methods

- **Predictive accuracy**
- **Speed and scalability**
 - Time to construct and use the model
 - Ability of handling large data set
- **Robustness**
 - Handling noise and missing values
- **Interpretability:**
 - understanding and insight provided by the model
- **Goodness of rules**
 - decision tree size
 - compactness of classification rules



Data preparation for classification

- **Data cleaning**

- Preprocess data in order to reduce noise and handle missing values (removing/imputation/interpolation)

- **Relevance analysis (feature selection)**

- Remove the irrelevant or redundant attributes

- **Data transformation**

- Generalize or normalize data

- Use discretization techniques to convert a given continuous attribute into categorical attribute, e.g., age, income, etc.

- Equal-width (distance) partitioning method

- Information based method, etc.

Decision Tree Induction

- **What is “induction”?**

- It is a reasoning approach where a concept can be learned/supported by gathering evidences from observations.

- **Decision Tree (DT):**

- In decision analysis, a **DT**(or tree diagram) is a decision support tool that uses a graph of tree, or model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility.

- All DT methods use DT as a hierarchical data structure and differ in how the tree is built (i.e., DT Induction).

- An internal node of DT associates with a selected attribute and arcs with values for that attribute.

Example: Decision Tree

- **Transportation mode choice:**

- Exercise at class (lecture: Class_note_classification_DT_example)

Example: Decision Tree

- Generate a model for Play Tennis:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rain	mild	high	weak	Yes
5	rain	cool	normal	weak	Yes
6	rain	cool	normal	strong	No
7	overcast	cool	normal	strong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rain	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	false	Yes
14	rain	mild	high	strong	No

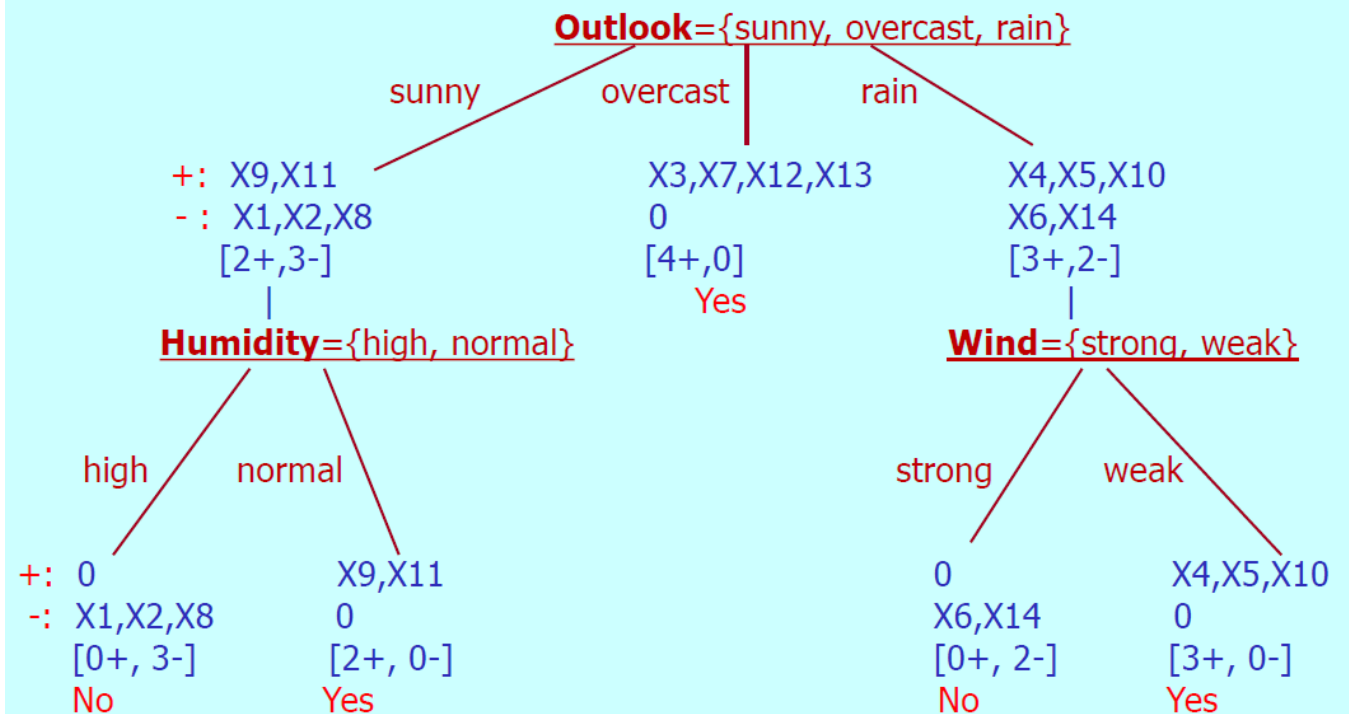
Example: Play Tennis

- Induction by sorting examples

D: (Outlook, Temperature, Humidity, Wind, PlayTennis)

+: X3,X4,X5,X7,X9,X11,X12,X13 Yes for PlayTennis

--: X1,X2,X6,X8,X14 No for PlayTennis

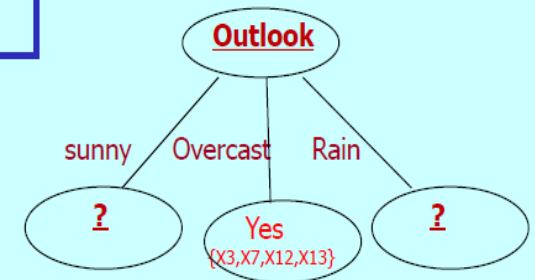


Example: Play Tennis

- DT: Splits Area: Sort examples into subareas
- Decision boundary
- Axis-parallel rectangle

$+: X9, X11$ $-: X1, X2, X8$	$+: X3, X7, X12, X13$ $-: 0$ Yes	$+: X4, X5, X10$ $-: X6, X14$
sunny	overcast	rain

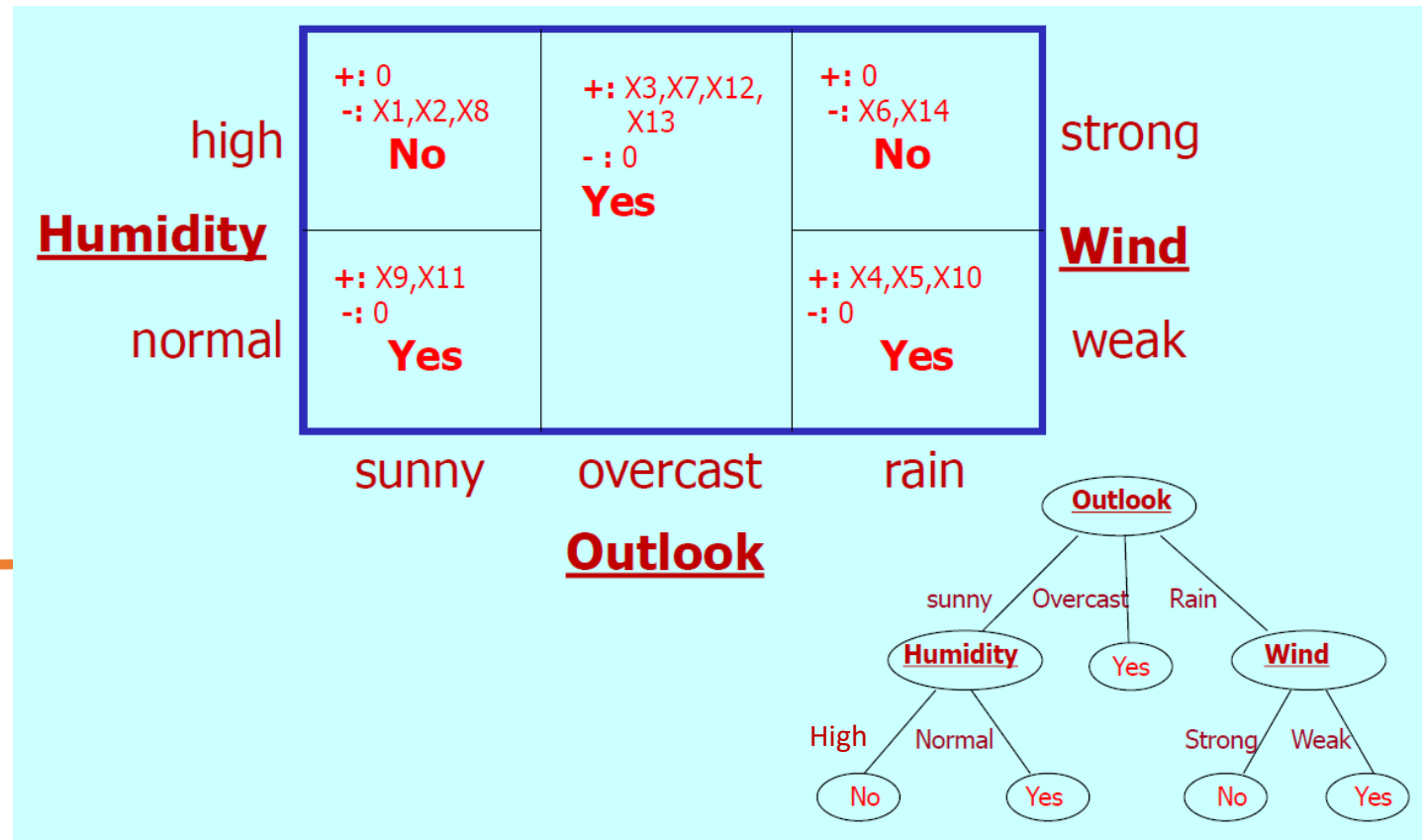
Outlook



D: (Outlook, Temperature, Humidity, Wind, PlayTennis)

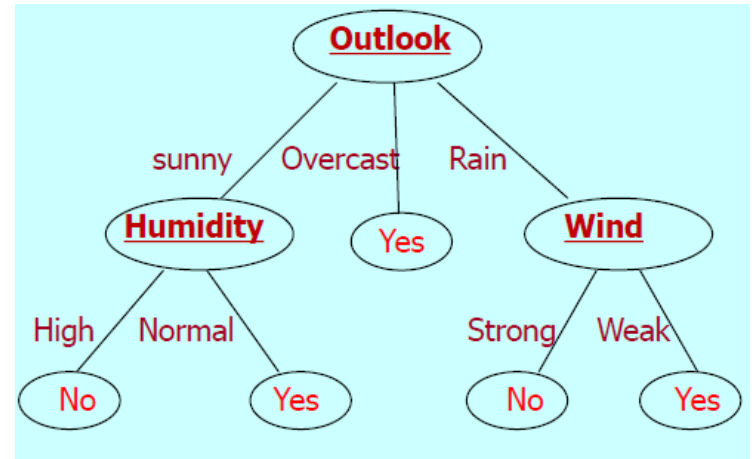
Example: Play Tennis

DT: Splits Area



Example: Play Tennis

DT: Splits Area

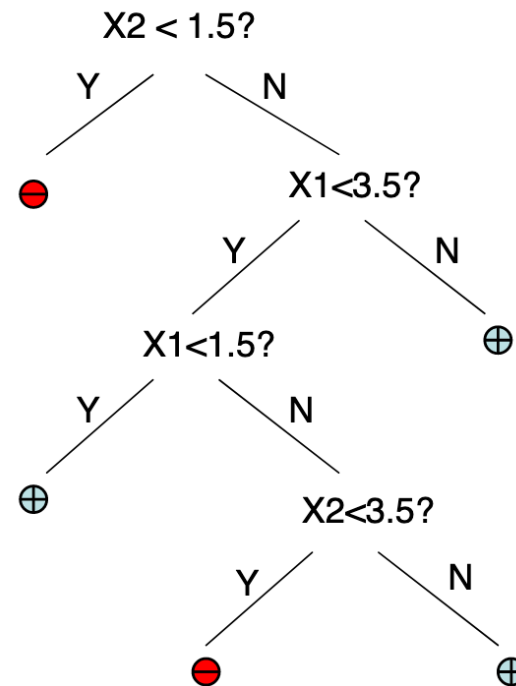
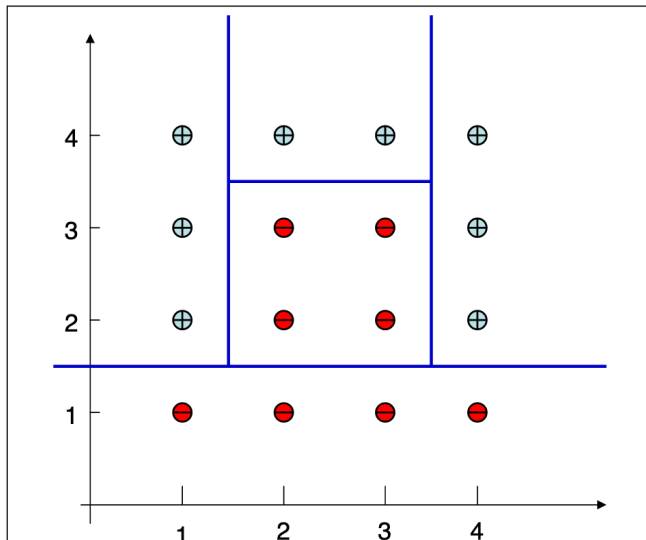


A decision table corresponding to the decision tree above. It is a 2x3 grid with 'Humidity' as the row header and 'Outlook' as the column header. The rows are labeled 'high' and 'normal' for Humidity. The columns are labeled 'sunny', 'overcast', and 'rain' for Outlook. The cells contain the predicted class: 'No' for (high, sunny), (high, rain), (normal, sunny), and (normal, rain); 'Yes' for (high, overcast) and (normal, overcast). A double-headed arrow points from the table to the decision tree.

<u>Humidity</u>	<u>Outlook</u>			<u>Wind</u>
	sunny	overcast	rain	
high	No	Yes	No	strong
normal	Yes	Yes	Yes	weak

Example: Decision Tree

- Decision boundaries?





Define DT

Given:

- Training data set D with n tuples: $D = \{t_1, \dots, t_n\}$
- Schema of D with m attributes: $(A_1, A_2, \dots, A_m)$
- Classes: $C = \{C_1, \dots, C_p\}$ of a selected target concept

Decision Tree (a tree associated with D) is defined:

- A **root node** has no incoming arcs and zero or more outgoing arcs
- Each **internal node** is a selected attribute, A_i
- Each **arc** is labeled with an attribute value (predicate condition) of the parent node
- Each **leaf node** is labeled with a class, C_j

Technique/Issues of DT

- 
-
- Preparing a training data set
 - Choosing an attribute to split
 - Splitting method
 - Control structure for tree construction
 - Stopping criteria
 - Pruning



DT Algorithms

- **ID3 (Iterative Dichotomiser 3)**

- Quinlan, J.R., "Induction of decision trees", Machine Learning, Vol. 1, No.1, pp 81-106, 1986.

- **C4.5 (C5.0)**

- Quinlan, J.R., "C4.5: Programs for Machine Learning", San Francisco: Morgan Kaufman, 1993.

- **Others:** CART, CHAID, Chi-squared etc.



DT Algorithms

- ID3, C4.5, and CART adopt a greedy (i.e., nonbacktracking) approach in which decision trees are constructed in a top-down recursive divide-and-conquer manner.
- **DT Induction Algorithm:** See Section 8.2.1
- **Main Issues**
 - Determine how to split the records
 - How to specify the attribute test condition?
 - How to determine the best split?
 - Determine when to stop splitting
 - Reach leaf node/Majority vote

How to specify test condition?



- **Depends on attribute types**

- Nominal/Categorical
- Ordinal
- Numeric/Continuous

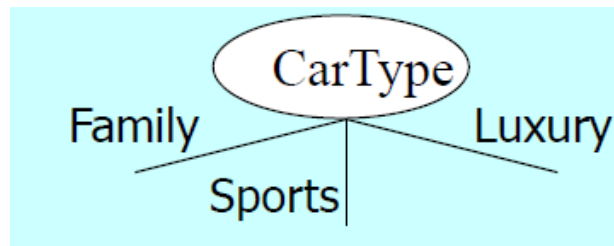
- **Depends on number of ways to split**

- 2-way split
- Multi-way split

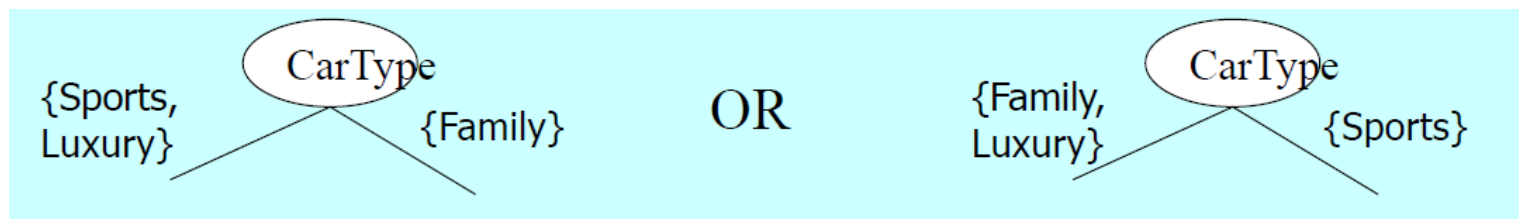


Splitting Based on Nominal Attributes

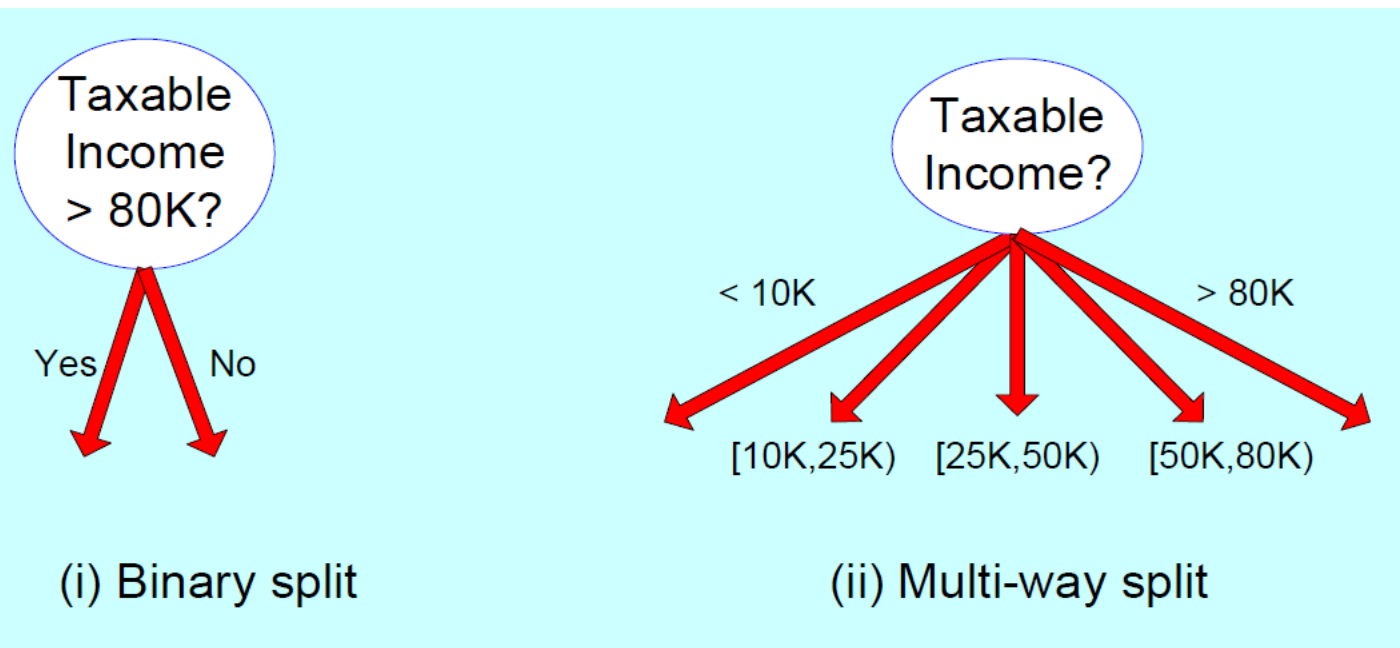
- **Multi-way split:** Use as many partitions as distinct values



- **Binary split:** Divides values into two subsets. Need to find optimal partitioning



Splitting Based on Continuous Attributes



How to determine the best split?

- **Which attribute is the best classifier?**
 - How well a given attribute split the records for classification
 - We want to select the best one to split
- **What is a good quantitative measure?**
 - **Information Gain** (a probability property)

(See lecture “Class_note_classification_DT_example” and Book Section 8.2.2)
- **ID3 uses information gain to select among the candidate attributes at each step while growing the tree**

How to measure information?

- How to measure the information contained in an attribute for classification of the target concept?
 - Information in general always associates with a question and a message for answering the question
 - Information measure is to quantify the outcome of some expectation from the message for answering the question

Information measure for classification: Class Purity

E.g., Let's look at classification for the concept PlayTennis for answer (i.e., class) "Yes":

- For expectation: PlayTennis = Yes
 - $P1_yes(before) = 9/14$
 - The probability of an unseen record being "Yes" class without asking any question.
 - Similarly, for "No" $P1_no(before) = 5/14$
 - For $P2_yes(after) = ?$
- Which question should be asked from {Outlook, Temperature, Humidity, Wind}?

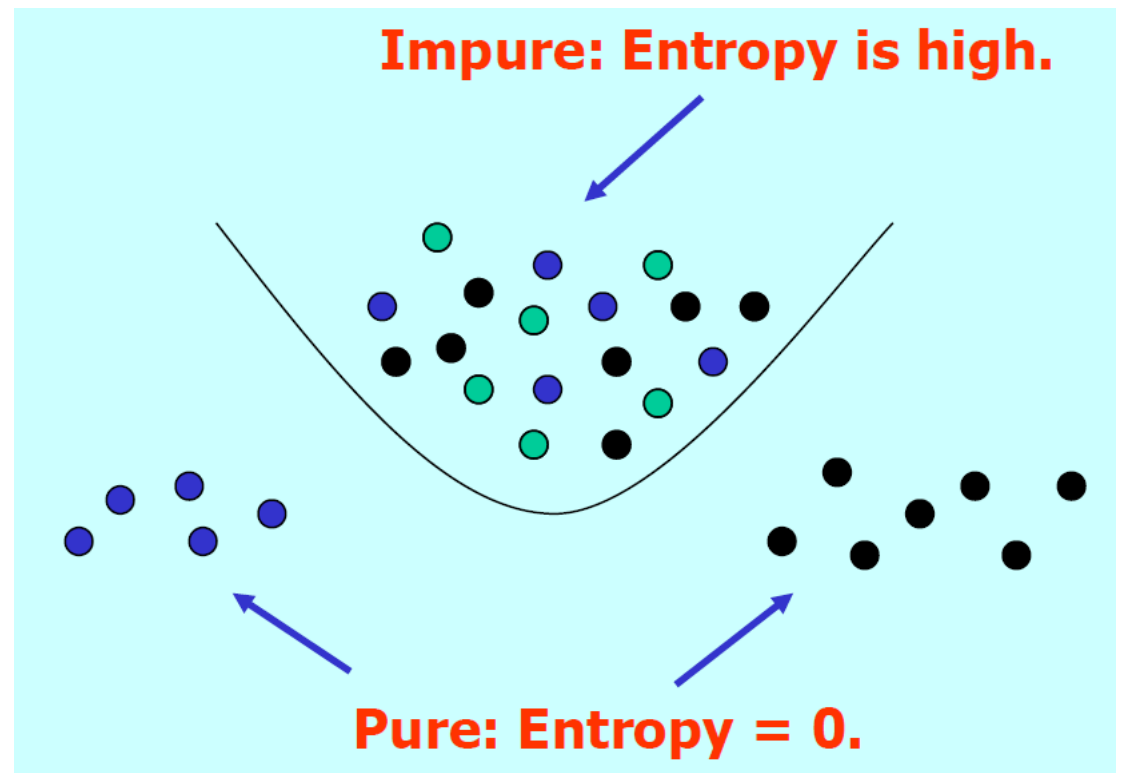


Which question should be asked from the given attributes (regarding class purity)?

- $P2_yes(after) = ?$
- The questions may be asked like "what is outlook?", or "what is Temperature?", etc. from the list of the attributes {Outlook, Temperature, Humidity, Wind}.
- E.g., What is Outlook?
 - Then the message can be from
$$P2_yes(Outlook=sunny) = 2/5,$$
$$P2_yes(Outlook= overcast) = 4/4,$$
$$P2_yes(Outlook= rain) = 3/5.$$
- $P2_no(after) = ?$
-
- How to quantify the information about containing both "Yes" and "No" classes? **Entropy.**

What is Entropy?

- In information theory
 - Entropy characterizes the impurity (or uncertainty) of an arbitrary set of data
 - For classification, entropy is used to estimate how close a data set to a single class.



Entropy: a measure of impurity of data

Homework!



- Bias and Variance/Overfitting and Underfitting
- Tree Pruning
- Read class note, chapter 2 (Section 2.1) and chapter 8 (mainly Section 8.1, 8.2, 8.3)

Thank you

